

● Original Contribution

SUBHARMONIC, NON-LINEAR FUNDAMENTAL AND ULTRAHARMONIC IMAGING OF MICROBUBBLE CONTRAST AT HIGH FREQUENCIES

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Abstract—There is increasing use of ultrasound contrast agent in high-frequency ultrasound imaging. However, conventional contrast detection methods perform poorly at high frequencies. We performed systematic *in vitro* comparisons of subharmonic, non-linear fundamental and ultraharmonic imaging for different depths and ultrasound contrast agent concentrations (Vevo 2100 system with MS250 probe and MicroMarker ultrasound contrast agent, VisualSonics, Toronto, ON, Canada). We investigated 4-, 6- and 10-cycle bursts at three power levels with the following pulse sequences: B-mode, amplitude modulation, pulse inversion and combined pulse inversion/amplitude modulation. The contrast-to-tissue (CTR) and contrast-to-artifact (CAR) ratios were calculated. At a depth of 8 mm, subharmonic pulse-inversion imaging performed the best (CTR = 26 dB, CAR = 18 dB) and at 16 mm, non-linear amplitude modulation imaging was the best contrast imaging method (CTR = 10 dB). Ultraharmonic imaging did not result in acceptable CTRs and CARs. The best candidates from the *in vitro* study were tested *in vivo* in chicken embryo and mouse models, and the results were in a good agreement with the *in vitro* findings. (E-mail: v.daeichin@erasmusmc.nl) © 2015 World Federation for Ultrasound in Medicine & Biology.

Key Words: Ultrasound contrast agent, Contrast imaging, High-frequency ultrasound, Chicken embryo, Contrast-to-tissue ratio, Non-linear propagation artifact, *In vitro*, *In vivo*.

INTRODUCTION

The need for high-resolution ultrasound imaging has increased the diagnostic use of high-frequency ultrasound (HFU, >15 MHz) (Brown et al. 2007; Foster et al. 2009, 2011; Lukacs et al. 2006; Needles et al. 2010; Ritter et al. 2002). In addition to its numerous applications in small animal imaging (Foster et al. 2011), HFU is widely used in the clinical diagnosis of intravascular, dermatologic and ophthalmologic pathology (Shung et al. 2009). Within the field of HFU, the use of ultrasound contrast agents (UCAs) is also growing (Foster et al. 2011; Needles et al. 2010). Although UCAs have been used and studied extensively at frequencies below 15 MHz (Wilson and Burns 2010), the behavior of UCA microbubbles above 15 MHz is not fully understood (Foster et al. 2011).

The stronger scattering behavior of UCAs is the key factor in linear contrast detection techniques (Ritter et al. 2002). However, these methods cannot provide sufficient contrast in images for many applications, such as detection of small capillaries in tissue perfusion or in the presence of tissue motion. Therefore, UCA-specific non-linear contrast imaging techniques are essential at high frequencies (>15 MHz). Conventional non-linear imaging techniques, at lower frequencies, focus mainly on detection of higher harmonics (Burns et al. 1994; Chang et al. 1995; de Jong et al. 2000; Deng and Lizzi 2002; Simpson et al. 1999). Similar techniques have been implemented at higher frequencies (Cachard et al. 1997; Lyshchik et al. 2007; Moran et al. 2002; Needles et al. 2010; Rychak et al. 2007; Willmann et al. 2008). However, the performance of these methods is degraded in HFU imaging because the driving frequency is much higher than the resonance frequency of the UCA. In addition, the propagation of the transmitted acoustic wave in tissue is more non-linear at higher frequencies (Blackstock 2000). Therefore, techniques exploiting higher harmonics are hampered because of the increase

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in the amplitude of harmonics produced by tissue (Goertz *et al.* 2005). These limitations motivate the exploitation of the non-linear components of the UCA response at lower frequencies: the non-linear fundamental (NF), sub-harmonic (SH) and ultraharmonic (UH) frequencies.

Contrast imaging focusing on the NF component of the UCA works best if the excitation frequency is close to the resonance frequency of the microbubbles. To obtain a strong enough NF signal from the currently used UCAs at frequencies above 15 MHz, high transmit pressures are required, which may disrupt the microbubbles. Another challenge in NF imaging is the so-called non-linear propagation artifact or far-wall artifact (Renaud *et al.* 2012; Tang *et al.* 2010; ten Kate *et al.* 2012; Thapar *et al.* 2012; Yu *et al.* 2010) produced by non-linear propagation of the excitation wave through the UCA which is backscattered by the tissue behind it, resulting in a false response at the site of the linearly scattering tissue.

Use of the emission of energy by the UCA at half the excitation frequency f_0 , the SH response (Eller and Flynn 1969), was proposed as a new imaging modality (Shankar *et al.* 1998). No SH signal is generated during propagation in tissue, and no SH scattering is produced by tissue. The backscattered SH signal is attenuated less in tissue than the signals at fundamental and higher harmonic frequencies, providing a potentially powerful diagnostic tool for clinical examinations (Forsberg *et al.* 2000). Nevertheless, the excitation bursts required for SH imaging should also be considered. The SH signal is strongly dependent on the applied acoustic pressures, the ambient pressure variations (Eller and Flynn 1969; Tjøtta and Tjøtta 1980) and the envelope of the excitation signal (Cosgrove and Lassau 2009; Daeichin *et al.* 2012). Previously, we reported that the self-demodulation (S-D) signal can enhance the SH response of phospholipid-coated microbubbles by up to 20 dB at 10 MHz (Daeichin *et al.* 2012). The S-D signal is a low-frequency signal component produced by weakly non-linear propagation of an ultrasound wave. It is proportional to the second time derivative of the squared envelope of the transmitted signal (Averkiou *et al.* 1993; Berkay 1965; Daeichin *et al.* 2012; Vos *et al.* 2010). Rectangular envelopes were found to produce strong SH stimulation of the UCA.

The UH response (at $3/2f_0$) constitutes another UCA-specific signal that is not generated by tissue (Maresca *et al.* 2013). UH contrast imaging can be performed with a low-bandwidth probe (40% bandwidth), whereas SH or higher harmonic imaging requires at least 70% bandwidth (Maresca *et al.* 2013). Also, the lateral resolution in UH imaging is higher than that in SH and NF imaging (Goertz *et al.* 2005). The drawbacks are the relatively weak level of the UH signal from the UCA

and the higher attenuation of the UH signal because of frequency-dependent attenuation.

In the study described here, we systematically compared, in an *in vitro* setup, the value of three different frequency components: SH, NF and UH. We used excitation bursts with a rectangular envelope together with conventional non-linear contrast detection methods: pulse inversion (PI) (Simpson *et al.* 1999), amplitude modulation (AM) (Brock-Fisher *et al.* 1996) and a combination of PI and AM (PIAM) (Haider and Chiao 1999). The comparison is based on measurements of the ratio of the detected UCA signal to the residual tissue signal (termed the *contrast-to-tissue ratio* [CTR]) and the ratio of the detected UCA signal to the far-wall artifact signal (termed the *contrast-to-artifact ratio* [CAR]). Target depth, pulse duration, UCA concentration and transmit pressure were systematically varied to find the optimal non-linear imaging approaches for HFU applications in different conditions. Finally, those combinations with the highest CTRs and CARs *in vitro* were chosen for *in vivo* verification in mouse and chicken embryo models.

The results described in this article are the primary findings of our comprehensive *systematic in vitro* analysis (108 measurements and 648 analyses in total). The results and details of the *in vivo* experiments are presented in the online supplement (available online at <http://www.umbjournal.org>), referred to as **Supplementary Material** throughout this article. All CTR and CAR values referred to in this article are highlighted and numbered from ①–⑥ in **Supplementary Tables 2–10** in the **Supplementary Material**.

METHODS

A high-frequency ultrasound scanner with linear array transducer (Vevo 2100 with MS250 probe, VisualSonics, Toronto, ON, Canada) was operated at three frequencies (15, 22 and 30 MHz). We insonified the UCA at transmit frequency f_0 and examined the frequency responses at around $1/2f_0$ for SH imaging, f_0 for NF imaging and $3/2f_0$ for UH imaging. The MS250 probe has a center frequency of 22.5 MHz and a –6-dB two-way frequency bandwidth of 70% (15–30 MHz) (Needles *et al.* 2010). To benefit from the transducer's sensitivity in both transmission and reception, the transmit frequencies were selected as follows: 15 MHz for UH imaging, resulting in a received UH component at 22.5 MHz; 22 MHz for NF imaging; and 30 MHz for SH imaging, with a received SH component at 15 MHz. Please note that in this study the terms SH and UH are defined with respect to the frequency of the transmitted ultrasound signal, not with respect to the resonance frequency of the UCA microbubbles.

In vitro measurements

A tissue-mimicking phantom (TMP) was prepared according to the recipe of [Teirlinck et al. \(1998\)](#). Two wall-less cylindrical cavities (5 mm in diameter) were created in two depths (center of the cylinders at 8 and 16 mm) by removing metallic rods from the TMP after the material solidified. MicroMarker contrast agent (VisualSonics) was reconstituted according to the manufacturer's instructions, resulting in a concentration of approximately 2×10^9 bubbles/mL, with a mean bubble diameter of $1.2 \mu\text{m}$ ([Needles et al. 2010](#)). Three dilutions of UCA suspensions in saline (1:500, 1:1000 and 1:10000) were prepared. The microbubble solutions were gently stirred with a magnetic stirrer to ensure homogeneous distribution of the UCA and then poured into the two cavities in the phantom while scanning. The ultrasound probe was mounted on the phantom in such a way that the center of the cylinder containing the UCA at a depth of 8 mm was at the focus of the transducer. We observed that there was no greater than a 2-dB change in the relative CTR at 16-mm depth when the focal depth was changed from 8 to 16 mm. Therefore, the focal point was kept at 8-mm depth throughout all experiments. Bursts with rectangular envelopes were transmitted with the pulse sequences AM, PI, PIAM and B-mode. The received echoes for each of the 108 combinations of pulse length (3), pressure (3), concen-

tration of microbubbles (3) and pulse sequence (4) were recorded as I/Q data. Radiofrequency (RF) data were reconstructed from the I/Q data of 10 consecutive frames (at a rate of 18 frames/s) and were analyzed off-line using MATLAB (Version R2012 b, The MathWorks, Natick, MA, USA). The RF data were digitally bandpass filtered around SH, UH and fundamental frequencies, with fifth-order Butterworth filters designed in MATLAB. The -6-dB frequency cutoffs of the filters were set to: 14–16 MHz for SH imaging, 21.5–23.5 MHz for UH imaging and 21–23 MHz for NF imaging. Five identical rectangular regions of interest in the TMP and UCA (*black boxes* in [Figure 1](#)) were selected. From the bandpass-filtered RF signals of the 10 frames recorded in the Fourier domain, mean echo powers and standard deviations were calculated for each region of interest and used to determine the CTR and CAR according to the equations.

$$\text{CTR at 8 - mm depth} = P_1/P_2 \quad (1)$$

$$\text{CTR at 16 - mm depth} = P_5/P_4 \quad (2)$$

$$\text{CAR} = P_1/P_3 \quad (3)$$

where P_i is the mean echo power in the region $i \in \{1 \dots 5\}$. [Figure 1](#) depicts the *in vitro* setup.

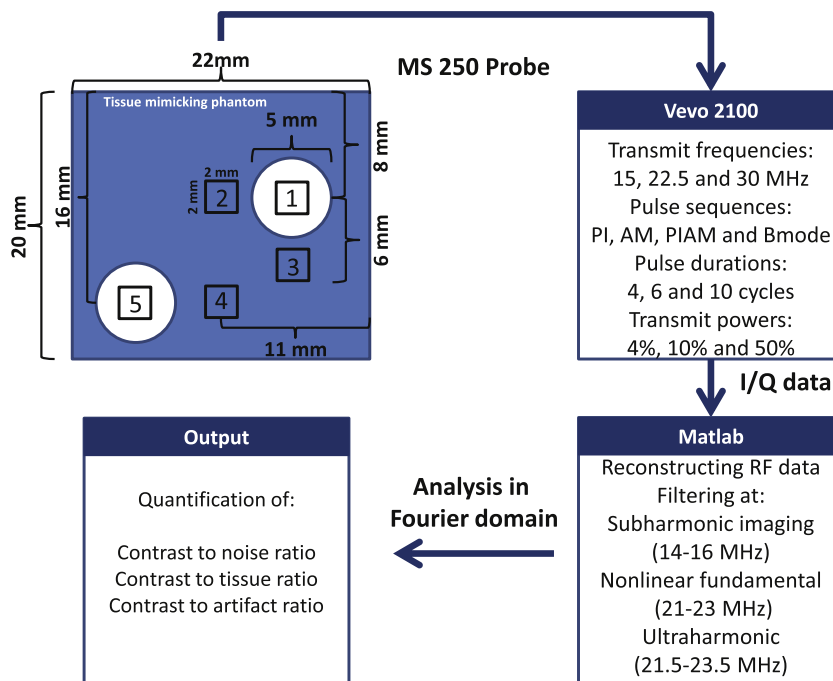


Fig. 1. Experimental setup for the *in vitro* characterization. Acoustic echoes were received with the MS250 array transducer, and I/Q sampled signals were exported from the Vevo 2100 system and processed off-line in MATLAB. Five regions of interest (*black rectangular boxes*) were selected in the ultrasound contrast agent and the tissue-mimicking material to quantify the contrast-to-tissue and contrast-to-artifact ratios.

For each of the transmit power levels, hydrophone measurements in a water tank were conducted to measure the actual acoustic pressure wave signals (Suppl. Figs. 1 and 2).

In vivo measurements

The best candidate combinations for non-linear contrast imaging with the highest CTR and CAR *in vitro* were applied in an *in vivo* validation in three chicken embryos and in a kidney of a healthy adult mouse. Chicken embryo is an example of a superficial and weakly attenuating target, whereas mouse kidney represents a deeper target with higher attenuation (see Supplementary Material for details on the preparation of these models).

Only SH imaging and NF imaging were tested *in vivo*, because UH imaging exhibited very low sensitivity. Real-time SH and 3-D SH imaging were conducted

with the PI sequence (see Supplementary Material for details on the imaging method). This pulse sequence had the maximum CTR *in vitro* at 8-mm depth (low attenuation) and, therefore, seemed to be a suitable method for contrast imaging in the chicken embryo model. The combination of NF imaging and AM sequence performed the best *in vitro* at a 16-mm depth (high attenuation); therefore, this combination was selected for contrast imaging in the mouse kidney model.

RESULTS

In vitro

In Figure 2 are SH, NF and UH phantom images acquired with the four different pulse sequences (AM, PI, PIAM and B-mode) for the most representative combination of parameters. Figure 3 illustrates the CTRs and CARs at depths of 8 and 16 mm for 10-cycle bursts at

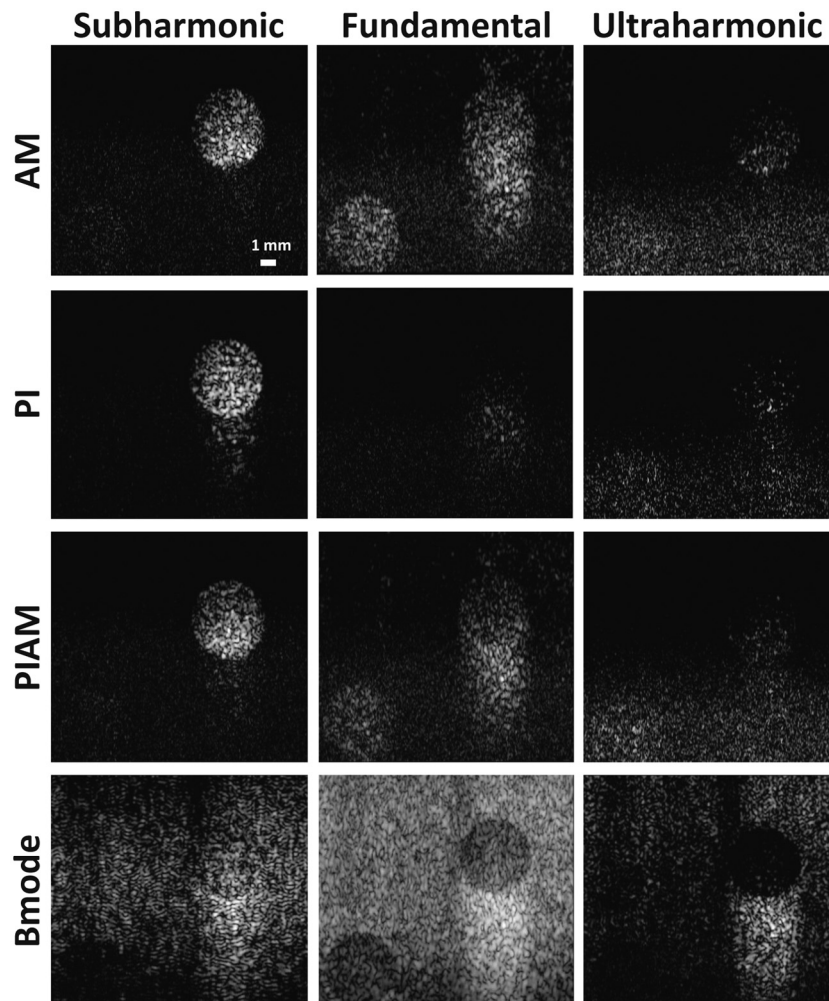


Fig. 2. *In vitro* subharmonic, non-linear fundamental and ultraharmonic images obtained with four pulse sequences (amplitude modulation, pulse inversion, combination of pulse inversion and amplitude modulation and B-mode) (1:1000 MicroMarker, 10% power, 10 cycles).

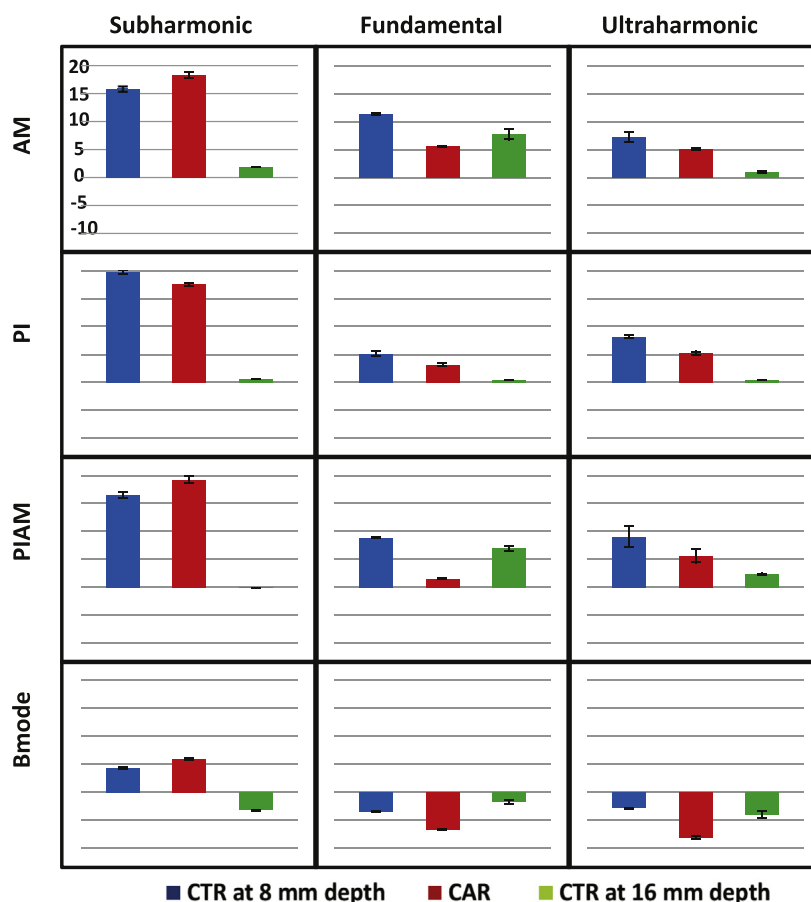


Fig. 3. Contrast-to-tissue ratios at depths of 8 and 16 mm and contrast-to-artifact ratios for subharmonic, non-linear fundamental and ultraharmonic images obtained with four pulse sequences (amplitude modulation, pulse inversion, combination of pulse inversion and amplitude modulation and B-mode). Error bars represent standard deviations. Transmit pressure was 10%, pulse duration was 10 cycles and MicroMarker dilution was 1:1000.

10% transmit power and a 1:1000 concentration of MicroMarker. The dynamic range for all images was 50 dB, and they were all normalized to a single reference value (maximum gray-scale value of the entire image set).

Subharmonic imaging. At an 8-mm depth, SH imaging performed much better than NF and UH imaging. The SH images had the highest CTRs at the 8-mm depth (blue bars in the first column of Fig. 3). Among the SH images, the highest CTR (20 dB) was achieved with the PI sequence (Fig. 3). The CTR for SH/PI/8 mm generally increased by increasing the burst length at 4% and 10% transmit power, with a maximum increase of 12 dB (Suppl. Table 5, ① and ④). Increasing the transmit power from 4% to 10% also generally increased the CTR for SH/PI/8 mm. Such enhancement in the CTR by the transmit power was a maximum of 7 dB (Suppl. Table 8, ③ and ④). However, a further increase in the transmit power to 50% significantly decreased the CTR (with a maximum drop of 18 dB [Suppl. Table 5, ④ and ⑥]).

At a depth of 16 mm, CTR values for SH imaging were significantly lower for all combinations. At 4% and 10% transmit power, the CTR values were <2 dB for all SH images. When the transmit power was increased to 50%, the CTR remained below 2 dB for SH images with the PI sequence.

Non-linear fundamental imaging. At a depth of 16 mm, NF was the best modality. The CTR of NF images at 16 mm reached 10 dB with the AM and PIAM sequences and medium transmit power (4% and 10%) (Suppl. Table 9, ③ and ④). The CTR decreased by as much as 9 dB in the NF images at higher depth (16 mm), but this drop in the CTR was considerably less than that in the SH images. However, at 8 mm, the highest CTR in the NF images was 8 dB lower than the highest CTR achieved in the SH images. The CTRs of the NF/AM/16 mm images increased by as much as 5 dB when the pulse length was increased to 10 cycles (Suppl. Table 9, ① and ③). Increasing the transmit power

in such a configuration did not improve the CTR more than 3 dB (Suppl. Table 9, ② and ③).

Ultraharmonic imaging. Ultraharmonic imaging was inferior to NF or SH imaging in any combination in our experiments. At a depth of 8 mm, the highest CTR was 16 dB for UH images with the PI sequence at 50% transmit power (Suppl. Table 8, ⑤). The CTR in UH images changed only marginally when the transmit power was increased from 4% to 10%. However, the CTR increased up to 9 dB when transmit power was further increased to 50% (Suppl. Table 5, ② and ③). No relevant CTR (<4 dB) was observed for the UH images at 16-mm depth, even at the highest transmit power (50%).

Artifact suppression. Subharmonic imaging suppressed the far-wall artifact the most. The SH images in Figure 3 had CAR values 11 and 8 dB higher than the CAR values of the NF and UH images. Variations in CAR values with transmit power and excitation length followed the same trends as observed for the CTR values (Suppl. Tables 4, 7 and 10). The CAR increased up to 9 dB for the SH/PI/8 mm images when the concentration of the UCA was increased from 1:10000 to 1:1000 (Suppl. Table 4, ①, and Suppl. Table 7, ①). No improvement in CAR was observed when the concentration was further increased to 1:500 for this configuration.

In vivo

According to our *in vitro* results, SH/PI images had the highest CTRs and CARs at 8-mm depth (low attenuation) and NF/AM images performed better at 16-mm depth (high attenuation). On the basis of these results, a series of *in vivo* experiments were conducted on chicken embryo (low attenuation) and mouse (high attenuation) models.

Chicken embryo model: Low attenuation. Figure 4 illustrates real-time SH/PI *in vivo* imaging of a 6-d chicken embryo. In Figure 4(a, b) are single frames in B-mode and SH-mode before injection of the UCA, and in Figure 4(c, d) are maximum-intensity projections of 480 frames after injection of the UCA. In the B-mode image, signals from the tissues around the vessels and heart hinder clear visualization of the UCA in the blood pool (Fig. 4c), therefore reducing the sensitivity and specificity of quantitative and qualitative analysis. In the SH image, the signals from the tissue are suppressed, and only the UCA is visualized (Fig. 4d). The trajectory of smaller vessels (about 100 μ m in diameter) and the three-chamber heart, as well as perfused surrounding tissue, are clearly visible in the SH image after injection of the UCA.

3-D subharmonic imaging. Figure 5 is a 3-D SH image of the heart and vasculature of an 8-d chicken embryo acquired after MicroMarker administration using the system's temporal filters. In the SH image, signals from the surrounding tissue were canceled, and only the microbubbles in the heart and vasculature around the heart were visualized.

As was suggested by Needles *et al.* (2010), at even higher frequencies (>24 MHz), SH imaging is preferred. In our study, an *in vivo* experiment was conducted in a 10-d chicken embryo using a higher-frequency probe (MS550s) with a transmit frequency of 40 MHz. The temporal filters of the system were used for real-time SH imaging by filtering the received echoes in the B-mode configuration. Three-dimensional images were acquired after injection of 5 μ L of MicroMarker (Fig. 6b). The enhancement of the three-chamber heart and the sub-millimeter vessels coming out of the heart can be appreciated.

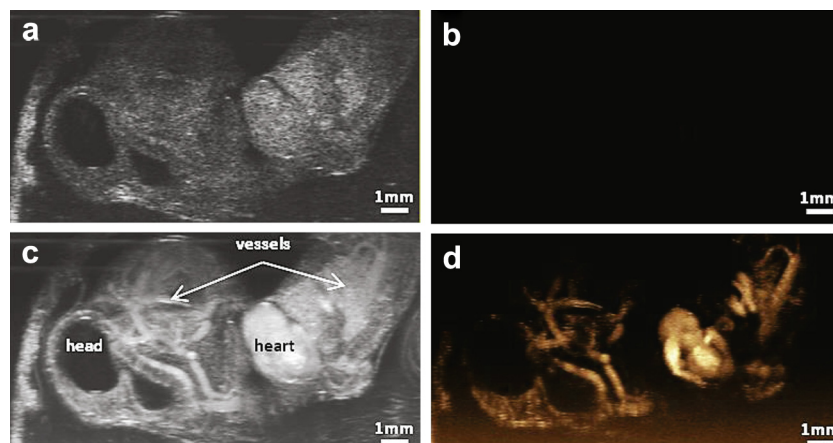


Fig. 4. B-Mode and real-time subharmonic images of a 6-d chicken embryo (a, b) before and (c, d) after injection of ultrasound contrast agent and application of a maximum-intensity projection on 480 frames (5 μ L MicroMarker, 10% power, 6 cycles).

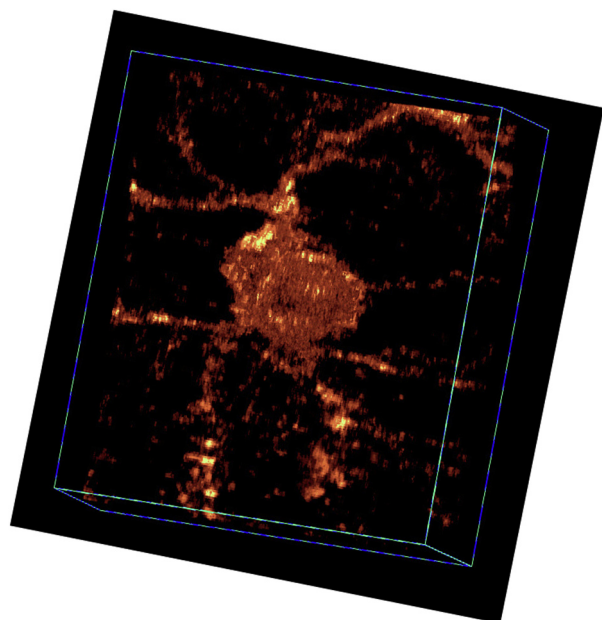


Fig. 5. Three-dimensional real-time subharmonic image of heart and vasculature of an 8-d chicken embryo after MicroMarker ultrasound contrast agent administration. The image was acquired with the Vevo 2100 system with MS250 probe using temporal filters and stepper motor (5 μ L MicroMarker, 6 cycles, 30 MHz, rectangular envelope, $\sim$ 300 kPa, rendered in Vevo 2100 3-D mode).

Mouse model: High attenuation. In Figure 7 are conventional B-mode, SH/PI and NF/AM images of a kidney of a mouse before and after injection of the UCA. In the B-mode image the microbubble signals were lost in the tissue signals in the kidney (Fig. 7d). The SH image obtained before injection of the UCA (Fig. 7b) had good tissue cancellation within the kidney, but the UCA was enhanced only minimally in the post-injection image (Fig. 7e). The NF images have good tissue cancellation (Fig. 7c), except for the top part of the image, which contains strong backscatter signals from the skin of the

animal. In the NF/AM image, the perfused kidney after injection of the UCA was very well visualized (Fig. 7f).

DISCUSSION

In vitro

Comparing the CTR measurements at depths of 8 and 16 mm in Figure 3 separately revealed that the best candidate for non-linear contrast imaging at low attenuation is SH/PI imaging, and at high attenuation, NF/AM imaging is the best. Figure 3 illustrates quantitative measures derived from the contrast images in Figure 2. In the following sections, each image modality is discussed in detail, and the effects of several imaging parameters are considered. The effects of UCA concentration and excitation pulse length on the imaging modalities are discussed in the Supplementary Material.

Subharmonic imaging. The subharmonic imaging mode provides the highest CTRs and CARs at 8-mm depth (Figs 2 and 3). CTR and CAR reached their maximum with the PI sequence, long excitation (10 cycles) and low transmit power (4%). One reason for this strong SH response of the UCA, even at a 30-MHz excitation frequency, could be the stimulation effect of the S-D signal at low transmit power. We had previously reported that a burst with rectangular envelope generates an S-D signal that enhances the SH response of the UCA (Daeichin et al. 2012). Although the transmit frequency for SH images in this work (30 MHz) is higher than that in the previous work (10 MHz), the S-D signal is still present and the stimulation effect is apparently effective in enhancement of the SH response of the UCA. It has been reported that the S-D signal is related to the second time derivative of the squared envelope of the excitation burst (Averkiou et al. 1993; Berkay 1965). Therefore, both excitation bursts in the PI sequence have an in-phase S-D signal, although the excitations at the fundamental frequency are in anti-phase. Such in-phase S-D

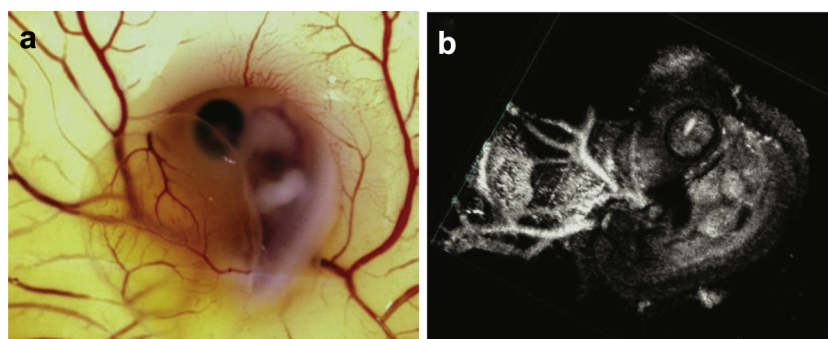


Fig. 6. (a) Optical image of a 10-d chicken embryo. (b) Three-dimensional subharmonic image of another 10-d chicken embryo after MicroMarker ultrasound contrast agent administration (5 μ L MicroMarker, 6 cycles, 40 MHz, rectangular envelope, $\sim$ 300 kPa, rendered in Vevo 2100 3-D mode).

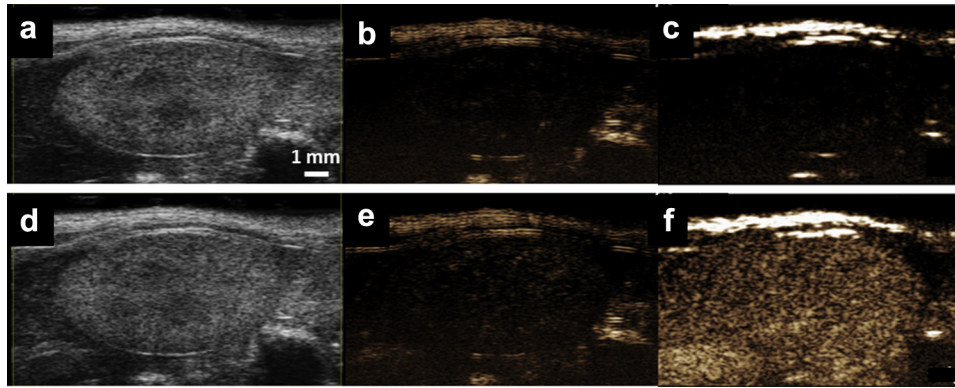


Fig. 7. B-Mode, subharmonic and non-linear fundamental images of a mouse kidney (a–c) before and (d–f) after injection of ultrasound contrast agent (100 μ L MicroMarker, 10% power, 6 cycles).

signals in the two transmit bursts can, in turn, trigger an in-phase SH response in the UCA. Consequently, the SH response of the UCA in the received echoes is enhanced when summation is applied in the PI sequence. Also, the PI sequence removes the linear fundamental components in the received echo signals. The spectral leakage of the linear fundamental components into SH frequencies is one of the main causes of image degradation in SH imaging and can be prevented by applying the PI sequence (Harput *et al.* 2013). Improvement of the CTR of SH/PI imaging has also been reported by Needles *et al.* (2010) at frequencies above 15 MHz. Furthermore, the narrowband excitation (10 cycles) improves the isolation of the SH component in the frequency domain. These effects can contribute to the improved CTR and CAR with the PI sequence and long excitations.

At 16-mm depth, the CTR decreased significantly (18 dB) (Fig. 3) for SH images. This is due mainly to the severe attenuation of the excitation bursts that propagate through the medium (Suppl. Figs. 1–3). Transmit signals for SH images were at 30 MHz, whereas transmit frequencies for NF and UH images were 22 and 15 MHz. Attenuation of the acoustic wave in the tissue-mimicking material is about 1 dB/MHz/cm of propagation (Sun *et al.* 2012). Therefore, the amplitude of the excitation burst at 30 MHz used for SH imaging is 2.5 times (8 dB) more attenuated than the bursts used for NF imaging (22 MHz). This highly attenuated burst may not be capable of stimulating the SH response of the UCA at a depth of 16 mm, especially because the SH response of the UCA exhibits threshold behavior with respect to the excitation amplitude (Eller and Flynn 1969; Shankar *et al.* 1999). Helfield *et al.* (2012) reported the pressure threshold for SH generation to be 119 kPa at 25 MHz for majority of MicroMarker in a native population, and Sprague *et al.* (2010) observed a thresholds of 110 kPa at 30 MHz for bound, targeted MicroMarker.

Subharmonic pulse-inversion imaging with a 30-MHz intravascular ultrasound transducer, which resulted in CTRs in the range 10–15 dB, was also reported by Goertz *et al.* (2006). Their results are in very good agreement with what we observed *in vitro*, noting that the image depth in their study was 1 to 6 mm (low depth). They also reported that at high frequencies, the CTR decreases dramatically as imaging depth and transmit pressure are increased. We argued that the reason for the decrease in CTR for SH imaging at higher depth is the severe attenuation of the excitation signal. Increasing the transmit power with the current system does not improve the CTR of the SH images at a 16-mm depth because the amplitude of the low-frequency ring-down and the S-D signals (around 15 MHz) increase significantly at higher transmit powers, whereas the amplitude of the pressure wave at the fundamental frequency (30 MHz) remains low (around 100 kPa) (Suppl. Fig. 3). The S-D and ring-down signals in the excitation burst have a component at around the SH frequency (15 MHz) that is backscattered by the tissue and degrades the CTR at high transmit powers for SH imaging. This undesired effect of the S-D and ring-down signal is discussed in more detail in the Supplementary Material.

Increasing the transmit power from 4% to 10% in our experiment increased the CTR for SH images at a depth of 8 mm, indicating that the SH response was in the growth regime at such transmit pressures (Eisenbrey *et al.* 2011). However the improvement in CTR did not continue when the power was raised to 50% because of the discussed effects of the high-amplitude S-D signal and ring-down signal of the transducer.

One solution to overcome such a limitation is to benefit from coded excitations, as suggested by Shekhar and Dooley (2012). Also, optimization of the envelope of the excitation bursts for less spectral leakage of the S-D signal into the SH frequencies or the use of a transducer with broader bandwidth can improve SH imaging

at high frequencies. Unfortunately with the imaging platform used in this study, such optimizations were not possible.

The CAR values are 12 dB higher in the SH images than in the NF images in [Figure 3](#). This is due to the fact that the SH signal is produced only by the UCA, and tissue does not exhibit any SH responses. Therefore, SH imaging is free from the artifact produced by the non-linear propagation of the acoustic wave through the UCA ([Renaud et al. 2011](#)).

Non-linear fundamental imaging. Our result suggests that with the current hardware used in this study, NF imaging is the only possibility for contrast imaging at a depth of 16 mm ([Fig. 3](#)). The *in vitro* results revealed that the CTR decreased only 3 dB between depths of 8 and 16 mm for the NF/AM images. NF imaging is less affected by attenuation, especially because there is no threshold behavior for the NF response of the UCA with respect to excitation pressure amplitude.

The NF components of the UCA response were best extracted with the AM sequence, resulting in the highest CTR and lowest artifact compared with other pulse sequences. Increasing the excitation burst length at a depth of 16 mm enhanced the CTR and CAR values in NF images. However, the 14-dB CTR with the shortest excitation ([Suppl. Table 8](#), ①) suggests that 4-cycle bursts are appropriate for NF imaging, providing the highest axial resolution.

The CTR of NF images is lower at a depth of 8 mm (e.g., 8 dB in [Fig. 3](#)) because of severe non-linear propagation and distortion at such high frequencies and high transmit power, especially when multipulse excitations of different amplitudes, as in the AM sequence, are used. Further reduction of transmit power to avoid such a problem would also suffer from electronic limitations of the transmitting and receiving hardware, such as imbalance in transistor end stages and diode offsets.

Ultra-harmonic imaging. Our *in vitro* results indicated the CTR was 12 dB lower for UH imaging than for SH imaging ([Fig. 3](#)). However, increasing the transmit power to 50% increased the CTR to 8 dB for UH imaging, whereas the CTR decreased for the SH and NF images. One reason for such enhancement of the CTR is the disruption of the UCA at such a high transmit pressure (0.8 MPa) and lower excitation frequency (15 MHz) (mechanical index = 0.2). Moreover, the low-frequency components of the S-D and ring-down signals in the transmit wave are absent or very weak at the UH frequency. Therefore, increasing the pressure will improve the CTR by increasing the amplitude of the UH response of the UCA with minimal increase in the tissue signal at UH frequency. The good performance of UH imaging at high transmit power is in agreement with the results of

[Maresca et al. \(2013\)](#), in which high-amplitude excitations were used (0.88 MPa). However, the CTR at low transmit pressure is much lower for UH imaging than for SH imaging, making UH less suitable for non-destructive non-linear contrast imaging. Further investigation is required to fully understand the UH behavior of the UCA at high frequencies.

In vivo

Chicken embryo model: Low attenuation. Using the temporal filters in the Vevo 2100 ultrasound system, we were able to perform real-time side-by-side B-mode and SH imaging. We also realized 3-D SH imaging by mechanically moving the MS250 2-D probe. The B-mode images give us the opportunity to assess the anatomic structure, whereas the functionality of the cardiovascular system can be assessed in the real-time SH images. Superimposing the SH images on the anatomic B-mode images in real time can also be beneficial for more precise monitoring during experiments. Advantages of side-by-side B-mode and SH imaging in real time and 3-D SH imaging have been reported previously at lower frequencies (transmit frequencies of 2.5 and 5.8 MHz) ([Eisenbrey et al. 2011, 2012](#); [Sridharan et al. 2013](#)). However, to the best of our knowledge, no such *in vivo* studies have been reported at frequencies above 15 MHz. [Needles et al. \(2010\)](#) performed SH imaging *in vitro* at high frequencies, but reported no *in vivo* results because of the inconsistency between their *in vitro* and *in vivo* experiments. We reported that in superficial *in vivo* targets such as the chicken embryo model, SH/PI imaging is feasible at transmit frequencies of 30 and 40 MHz.

The good performance of high-frequency SH imaging for superficial targets allows us to introduce the chicken embryo as a useful model for studying cardiovascular parameters and diseases. Ultrasound imaging of such a model has been reported before ([Fechner et al. 2008](#); [Schellpfeffer and Kolesari 2012](#); [Schellpfeffer et al. 2005](#)), but with a very poor image resolution and sensitivity. Moreover, to our knowledge, there is no study indicating the possibility of real-time non-linear contrast imaging at high frequencies in this model. We found that sensitive SH imaging can be performed in this model at frequencies up to 40 MHz, providing an opportunity to assess the anatomic structure and physiologic functions of the cardiovascular system of the chicken embryo model ([Figs. 4–6](#)).

Mouse model: High attenuation. *In vivo* contrast images of mouse kidney ([Fig. 7](#)) as a target with higher attenuation than the chicken embryo model are in agreement with our *in vitro* results. SH/PI imaging had a poor CTR in the kidney, and the NF/AM image

revealed the excellent performance of this real-time high-frequency contrast imaging mode for such *in vivo* conditions. The targets in the mouse model are deeper than those in the chicken embryo model. In addition, the skin of the mouse reflects a large amount of excitation acoustic signal (bright layer in Fig. 7). Therefore, the excitation signal experienced by the UCA deeper in the imaged area is considerably attenuated. This large attenuation of the transmitted signal reduces the SH response of the UCA. This attenuation effect is less challenging for NF imaging, as is seen in the NF images (Fig. 7f).

Calculating the perfusion parameters from time–intensity curves (TICs) in the SH and NF images (see [Supplementary Material](#) for details) revealed that NF imaging is more sensitive than SH imaging in estimation of the relative blood volume of the kidney. The relative blood volume is estimated from the peak intensity in the TIC, which can be estimated with more sensitivity by NF. This is in agreement with our *in vitro* results, which revealed that the SH response of the UCA at a depth of 16 mm is weaker than the NF response. The measurement of blood velocity, however, is dependent only on the speed at which the UCA enters the imaging plane and is independent of the concentration (intensity in the contrast image) (Arditi *et al.* 2006). The velocity measurements from the slope of the normalized TIC were the same for NF and SH images. This suggests that although SH imaging is less sensitive than NF imaging, it can still be used to derive perfusion parameters at great depths.

Limitations

Our results in this study were derived with a particular ultrasound system, transducer and UCA type. Although we expect that the results are applicable more generally, this has to be confirmed. This study was explorative in nature and investigated a large set of parameter combinations and circumstances to determine promising combinations. We could not explore certain promising parameter combinations (*e.g.*, lower power levels) because of the practical limitations of the hardware. The respective findings should be investigated further in more detail, with a smaller number of interesting parameters, over a wider range. Also, the statistical value of the differences found can be improved by large-scale repetitions of the most important experiments. Because of the huge number of parameter combinations investigated (648) we had to severely limit the number of repetitions per experiment. Additionally, it should be further investigated how these results can be translated into *in vivo* applications. Future work should focus on optimizing the imaging parameters *in vivo* and on advancements in the technology of imaging systems, transducers and UCAs

to improve on the results reported here. Extrapolating the findings of this study to even higher frequencies should be done with more caution. For frequencies >80 MHz, the microbubble radius can no longer be assumed to be small compared with the wavelength of the excitation wave. Therefore, the formulations for a multitude of well-known parameters used in the acoustics of UCA can no longer be trusted.

Practical implications

Our results illustrate that with the most frequently used HFU platform for small animal imaging (Vevo 2100, MS250 probe), improvements in high-frequency UCA detection are achievable with the SH/PI combination for superficial targets and the NF/AM combination for deeper targets. Examples of *in vivo* images of chicken embryo and mouse models confirm the promise for non-linear contrast imaging at high frequencies.

Further optimization of the pulse sequences and transducer response for SH/PI at higher transmit power and better implementation of the AM sequence at low transmit power for NF imaging can significantly improve the sensitivity of SH/PI and NF/AM for HFU contrast imaging.

We found that the suppression of the far-wall artifact with SH/PI imaging (CAR of 17 dB in Fig. 3) is an important feature of this imaging modality. Although, to the best of our knowledge, the effect of this artifact on HFU contrast imaging has not been reported, our *in vitro* results confirm the existence of the artifact, especially for the NF imaging at high frequencies.

The length of the excitation bursts is directly affecting the axial resolution in the ultrasound images. Although long narrowband excitations have higher CTRs for SH and NF imaging, our results indicated that the 6-cycle bursts for SH imaging and 4-cycle bursts for NF imaging are good compromises between high CTR values (up to 19 and 14 dB) and high axial resolution both *in vitro* (Suppl. Table 8, ① and ②) and *in vivo* (Figs. 4–7).

CONCLUSIONS

In our *in vitro* results, subharmonic imaging performed the best in media with low attenuation. Non-linear fundamental imaging is a more suitable approach for deeper and more attenuating media. The largest contrast-to-tissue and contrast-to-artifact ratios for the subharmonic images are achieved when pulse inversion is applied, and for the non-linear fundamental images, when amplitude modulation is used. To benefit from the advantages of the subharmonic signal as an imaging modality for deeper targets, hardware improvements are needed.

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SUPPLEMENTARY DATA

Supplementary data related to this article can be found online at <http://dx.doi.org/10.1016/j.ultrasmedbio.2014.10.003>.

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